

Backpropagation of layer normalization

Here we talk about the 2D situation where the shape of a batch of data is two-dimensional: the first dimension is batch size, and the second dimension is the length of the feature of a data sample.

We assume:

- The input of layer normalization:

$$X = \underbrace{\begin{bmatrix} -d_1 & - \\ -d_2 & - \\ \vdots & \end{bmatrix}}_d \Bigg\} BS$$

- The output of layer normalization(Before affine transformation):

$$Z = \underbrace{\begin{bmatrix} -d_1 & - \\ -d_2 & - \\ \vdots & \end{bmatrix}}_d$$

Final output after affine transformation:

$$O = \vec{\mathbf{w}} \odot Z + \vec{b}$$

Partial derivative of loss with respect to Z :

$$\frac{\partial L}{\partial Z} = \frac{\partial L}{\partial O} \frac{\partial O}{\partial Z} = \frac{\partial L}{\partial O} \vec{\mathbf{w}}$$

We know:

$$Z = \frac{X - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

Let's assume we have such input:

$$X = \begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \end{bmatrix}$$

The corresponding output is:

$$Z = \begin{bmatrix} z_1 & z_2 & z_3 \\ z_4 & z_5 & z_6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{x_1 - \mu(x_1, x_2, x_3)}{\sqrt{\sigma^2(x_1, x_2, x_3) + \epsilon}} & \frac{x_2 - \mu(x_1, x_2, x_3)}{\sqrt{\sigma^2(x_1, x_2, x_3) + \epsilon}} & \frac{x_3 - \mu(x_1, x_2, x_3)}{\sqrt{\sigma^2(x_1, x_2, x_3) + \epsilon}} \\ \frac{x_4 - \mu(x_4, x_5, x_6)}{\sqrt{\sigma^2(x_4, x_5, x_6) + \epsilon}} & \frac{x_5 - \mu(x_4, x_5, x_6)}{\sqrt{\sigma^2(x_4, x_5, x_6) + \epsilon}} & \frac{x_6 - \mu(x_4, x_5, x_6)}{\sqrt{\sigma^2(x_4, x_5, x_6) + \epsilon}} \end{bmatrix}$$

, and the partial derivative is a Jacobian matrix:

$$\frac{\partial Z}{\partial X} = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \frac{\partial z_1}{\partial x_3} & 0 & 0 & 0 \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \frac{\partial z_2}{\partial x_3} & 0 & 0 & 0 \\ \frac{\partial z_3}{\partial x_1} & \frac{\partial z_3}{\partial x_2} & \frac{\partial z_3}{\partial x_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\partial z_4}{\partial x_4} & \frac{\partial z_4}{\partial x_5} & \frac{\partial z_4}{\partial x_6} \\ 0 & 0 & 0 & \frac{\partial z_5}{\partial x_4} & \frac{\partial z_5}{\partial x_5} & \frac{\partial z_5}{\partial x_6} \\ 0 & 0 & 0 & \frac{\partial z_6}{\partial x_4} & \frac{\partial z_6}{\partial x_5} & \frac{\partial z_6}{\partial x_6} \end{bmatrix}$$

We can classify the partial derivative as two types:

$$D = \begin{cases} \frac{\partial z_i}{\partial x_i} \\ \frac{\partial z_i}{\partial x_j}, i \neq j \end{cases}$$

The definition of z_i is:

$$z_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$$

Here are some utils:

$$\begin{aligned}
\frac{\partial \mu}{\partial x_i} &= \frac{1}{d} \\
\frac{\partial \sqrt{\sigma^2 + \epsilon}}{\partial x_i} &= \frac{\partial \sqrt{\sigma^2 + \epsilon}}{\partial (\sigma^2 + \epsilon)} \frac{\partial (\sigma^2 + \epsilon)}{\partial x_i} \\
&= \frac{1}{2\sqrt{\sigma^2 + \epsilon}} \frac{\partial \left(\frac{1}{d} \sum_{j=0}^d (x_j - \mu)^2 + \epsilon \right)}{\partial x_i} \\
&= \frac{1}{2\sqrt{\sigma^2 + \epsilon}} \frac{\partial \left(\frac{1}{d} \sum_{j=0}^d (x_j - \mu)^2 \right)}{\partial x_i} \\
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}} \left(\frac{\partial \left(\sum_{j=0, j \neq i}^d (x_j - \mu)^2 \right)}{\partial x_i} + \frac{\partial ((x_i - \mu)^2)}{\partial x_i} \right) \\
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}} \left(\frac{\partial \left(\sum_{j=0, j \neq i}^d (x_j - \mu)^2 \right)}{\partial (x_i - \mu)} \frac{\partial (x_i - \mu)}{\partial x_i} + \frac{\partial ((x_i - \mu)^2)}{\partial x_i} \right) \\
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}} \left(\sum_{j=0, j \neq i}^d \frac{\partial ((x_j - \mu)^2)}{\partial (x_j - \mu)} \frac{\partial (x_j - \mu)}{\partial x_i} + \frac{\partial ((x_i - \mu)^2)}{\partial (x_i - \mu)} \frac{\partial (x_i - \mu)}{\partial x_i} \right) \\
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}} \left(\sum_{j=0, j \neq i}^d 2(x_j - \mu) \left(-\frac{1}{d} \right) + 2(x_i - \mu) \times \left(1 - \frac{1}{d} \right) \right) \\
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}} \left(\left(-\frac{1}{d} \right) \left(\sum_{j=0, j \neq i}^d 2(x_j - \mu) + 2(x_i - \mu) \right) + 2(x_i - \mu) \right) \\
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}} \left(\left(-\frac{1}{d} \right) \left(\sum_{i=0}^d 2(x_i - \mu) \right) + 2(x_i - \mu) \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2d\sqrt{\sigma^2 + \epsilon}}(0 + 2(x_i - \mu)) \\
&= \frac{x_i - \mu}{d\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{z_i}{d}
\end{aligned}$$

Let's look at these two situations separately:

$$\begin{aligned}
\frac{\partial z_i}{\partial x_i} &= \frac{\partial}{\partial x_i} \left(\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \right) \\
&= \frac{(1 - \frac{1}{d})\sqrt{\sigma^2 + \epsilon} - (x_i - \mu)(\sqrt{\sigma^2 + \epsilon})'}{\sigma^2 + \epsilon} \\
&= \frac{(1 - \frac{1}{d})\sqrt{\sigma^2 + \epsilon} - (x_i - \mu)\frac{z_i}{d}}{\sigma^2 + \epsilon} \\
&= \frac{(1 - \frac{1}{d}) - \frac{(x_i - \mu)}{\sqrt{\sigma^2 + \epsilon}} \frac{z_i}{d}}{\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{(1 - \frac{1}{d}) - \frac{z_i^2}{d}}{\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{d - 1 - z_i^2}{d\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{1 - \frac{1}{d} - \frac{z_i^2}{d}}{\sqrt{\sigma^2 + \epsilon}}
\end{aligned}$$

$$\frac{\partial z_i}{\partial x_j} = \frac{\partial}{\partial x_j} \left(\frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \right)$$

$$\begin{aligned}
&= \frac{(-\frac{1}{d})\sqrt{\sigma^2 + \epsilon} - (x_i - \mu)(\sqrt{\sigma^2 + \epsilon})'}{\sigma^2 + \epsilon} \\
&= \frac{(-\frac{1}{d})\sqrt{\sigma^2 + \epsilon} - (x_i - \mu)\frac{z_j}{d}}{\sigma^2 + \epsilon} \\
&= \frac{(-\frac{1}{d}) - \frac{(x_i - \mu)}{\sqrt{\sigma^2 + \epsilon}} \frac{z_j}{d}}{\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{(-\frac{1}{d}) - \frac{z_i z_j}{d}}{\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{-\frac{1}{d} - \frac{z_i z_j}{d}}{\sqrt{\sigma^2 + \epsilon}}
\end{aligned}$$

We can combine the two expression above by:

$$\begin{aligned}
\frac{\partial z_i}{\partial x_j} &= \frac{\delta_{ij} - \frac{1}{d} - \frac{z_i z_j}{d}}{\sqrt{\sigma^2 + \epsilon}} \\
\delta_{ij} &= \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}
\end{aligned}$$

We know:

$$\frac{\partial O}{\partial Z} = \begin{bmatrix} \frac{\partial O_1}{\partial z_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\partial O_2}{\partial z_2} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\partial O_3}{\partial z_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\partial O_4}{\partial z_4} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\partial O_5}{\partial z_5} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\partial O_6}{\partial z_6} \end{bmatrix}$$

we can use similar thinking to know:

$$\frac{\partial L}{\partial O} = \begin{bmatrix} \frac{\partial L}{\partial O_1} \\ \frac{\partial L}{\partial O_2} \\ \frac{\partial L}{\partial O_3} \\ \frac{\partial L}{\partial O_4} \\ \frac{\partial L}{\partial O_5} \\ \frac{\partial L}{\partial O_6} \end{bmatrix}$$

and we know:

$$\frac{\partial Z}{\partial X} = \begin{bmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \frac{\partial z_1}{\partial x_3} & 0 & 0 & 0 \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \frac{\partial z_2}{\partial x_3} & 0 & 0 & 0 \\ \frac{\partial z_3}{\partial x_1} & \frac{\partial z_3}{\partial x_2} & \frac{\partial z_3}{\partial x_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\partial z_4}{\partial x_4} & \frac{\partial z_4}{\partial x_5} & \frac{\partial z_4}{\partial x_6} \\ 0 & 0 & 0 & \frac{\partial z_5}{\partial x_4} & \frac{\partial z_5}{\partial x_5} & \frac{\partial z_5}{\partial x_6} \\ 0 & 0 & 0 & \frac{\partial z_6}{\partial x_4} & \frac{\partial z_6}{\partial x_5} & \frac{\partial z_6}{\partial x_6} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1 - \frac{1}{d} - \frac{z_1^2}{d}}{\sqrt{\sigma^2 + \epsilon}} & \frac{-\frac{1}{d} - \frac{z_2 z_1}{d}}{\sqrt{\sigma^2 + \epsilon}} & \frac{\partial z_1}{\partial x_3} & 0 & 0 & 0 \\ \frac{-\frac{1}{d} - \frac{z_1 z_2}{d}}{\sqrt{\sigma^2 + \epsilon}} & \frac{1 - \frac{1}{d} - \frac{z_2^2}{d}}{\sqrt{\sigma^2 + \epsilon}} & \frac{\partial z_2}{\partial x_3} & 0 & 0 & 0 \\ \frac{-\frac{1}{d} - \frac{z_1 z_3}{d}}{\sqrt{\sigma^2 + \epsilon}} & \frac{-\frac{1}{d} - \frac{z_2 z_3}{d}}{\sqrt{\sigma^2 + \epsilon}} & \frac{\partial z_3}{\partial x_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\partial z_4}{\partial x_4} & \frac{\partial z_4}{\partial x_5} & \frac{\partial z_4}{\partial x_6} \\ 0 & 0 & 0 & \frac{\partial z_5}{\partial x_4} & \frac{\partial z_5}{\partial x_5} & \frac{\partial z_5}{\partial x_6} \\ 0 & 0 & 0 & \frac{\partial z_6}{\partial x_4} & \frac{\partial z_6}{\partial x_5} & \frac{\partial z_6}{\partial x_6} \end{bmatrix}$$

so:

$$\frac{\partial L}{\partial X} = \left(\frac{\partial L}{\partial O} \right)^T \frac{\partial O}{\partial Z} \frac{\partial Z}{\partial X}$$

We set:

$$\frac{\partial L}{\partial O_i} = g_i$$

For a certain term:

$$\begin{aligned} \frac{\partial L}{\partial X_{11}} &= \frac{\partial L}{\partial x_1} \\ &= \sum_{j=1}^d g_j w_j \frac{\delta_{j1} - \frac{1}{d} - \frac{z_1 z_j}{d}}{\sqrt{\sigma^2 + \epsilon}} \\ &= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \sum_{j=1}^d g_j w_j \left(\delta_{j1} - \frac{1}{d} - \frac{z_1 z_j}{d} \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \left(\sum_{j=1}^d g_j w_j \delta_{j1} - \frac{1}{d} \sum_{j=1}^d g_j w_j - \frac{z_1}{d} \sum_{j=1}^d g_j w_j z_j \right) \\
&= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \left(g_1 w_1 - \frac{1}{d} \sum_{j=1}^d g_j w_j - \frac{z_1}{d} \sum_{j=1}^d g_j w_j z_j \right)
\end{aligned}$$

$$\begin{aligned}
\frac{\partial L}{\partial X_{12}} &= \sum_{j=1}^d g_j w_j \frac{\delta_{j2} - \frac{1}{d} - \frac{z_2 z_j}{d}}{\sqrt{\sigma^2 + \epsilon}} \\
&= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \sum_{j=1}^d g_j w_j \left(\delta_{j2} - \frac{1}{d} - \frac{z_2 z_j}{d} \right) \\
&= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \left(\sum_{j=1}^d g_j w_j \delta_{j2} - \frac{1}{d} \sum_{j=1}^d g_j w_j - \frac{z_2}{d} \sum_{j=1}^d g_j w_j z_j \right) \\
&= \frac{1}{\sqrt{\sigma^2 + \epsilon}} \left(g_2 w_2 - \frac{1}{d} \sum_{j=1}^d g_j w_j - \frac{z_2}{d} \sum_{j=1}^d g_j w_j z_j \right)
\end{aligned}$$

$$\frac{\partial L}{\partial X} = \frac{1}{\sqrt{\sigma^2 + \epsilon}} \left(G \odot \vec{w} - \text{mean}(G \odot \vec{w}) - Z \odot \text{mean}(G \odot \vec{w} \odot Z) \right)$$

where,

$$\begin{aligned}
\text{mean}(G \odot \vec{w}) &= \begin{bmatrix} \text{mean}(G[0] \odot \vec{w}) \\ \text{mean}(G[1] \odot \vec{w}) \\ \vdots \\ \text{mean}(G[BS-1] \odot \vec{w}) \end{bmatrix} \\
\text{mean}(G \odot \vec{w} \odot Z) &= \begin{bmatrix} \text{mean}(G[0] \odot \vec{w} \odot Z[0]) \\ \text{mean}(G[1] \odot \vec{w} \odot Z[1]) \\ \vdots \\ \text{mean}(G[BS-1] \odot \vec{w} \odot Z[BS-1]) \end{bmatrix}
\end{aligned}$$

So, in code implementation, we can write:

```
1 # delta: bs x d
2 # Z: bs x d
3 # X: bs x d
4 L20 = delta
5 02Z = w
6
7 L2X = 1/torch.sqrt(X.var(dim=1, keepdims=True,
8 correction=0)+epsilon) *
9 (L20 * 02Z -
10 (L20 * 02Z).mean(dim=1, keepdims=True) -
11 Z * (L20*02Z*Z).mean(dim=1, keepdims=True))
```